
Leveraging Synthetic Data for Speech Recognition: Improving Shunting Commandos with Conformer-CTC at Swiss Federal Railways

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Abstract

In this study, we investigated the potential of synthetic data augmentation to boost the performance of automatic speech recognition (ASR) models, specifically within the context of the Swiss Federal Railways (SBB). Utilizing Azure's text-to-speech service, we created a synthetic dataset that simulates real-world railway commandos, effectively expanding the diversity and size of our training data. We employed the Conformer-CTC model and trained it on various blends of human-recorded and synthetic data, assessing its performance by using the Word Error Rate (WER). Our results demonstrated a significant reduction in WER when the model was trained on a combination of human and synthetic data, with the most significant improvements occurring when the training set consisted of 25% human recorded and 75% synthetic generated samples (total 8469 Samples). However, the model's performance plateaued beyond certain levels of synthetic data inclusion, indicating a nuanced relationship between the amount of synthetic data used and the model performance. This research highlights the transformative potential of synthetic data augmentation in ASR tasks and opens avenues for further exploration of more sophisticated synthetic data generation and utilization techniques to enhance the performance of ASR systems.

1 Introduction

In the rapidly advancing world of AI, automation has proven to be a pivotal factor in enhancing operational efficiency and accuracy, transcending various industry boundaries. The Swiss Federal Railways (SBB), handling over 700,000 shunting orders annually, stands at the crossroads of this transformative journey. Currently, their process relies on a manual system where employees communicate verbally via smartphones, and instructions are transcribed by office staff—a method not devoid of its inherent challenges. To enhance comprehension, the following constitutes an example of shunting communication at SBB:

"Rangierfahrt von Bravo Eins nach Gleis Hotel Fünf"

Among the primary obstacles is the diversity in voices and accents, attributable to SBB's multicultural workforce, which includes speakers of various languages like German, French, and Italian, each with their own unique accents. This complicates the task of accurately recognizing speech, especially considering the phonetic proximity of certain German words used in communications, like "via" and "vier". Further, the scarcity of training and testing samples—limited to around 3500 samples hinders the development of a robust and generalizable speech-to-text model.

In response to these challenges, SBB outlined several goals for this project. The primary objective is the creation of a comprehensive dataset comprising approximately 20,000 training examples. This ensures the model is exposed to a broad spectrum of scenarios, improving its generalizability. A crucial aspect of this goal is the inclusion of examples with accents, aiming to enhance the model's adaptability to diverse language variations. The project also emphasizes the utilization of the Conformer-CTC model, a potent architecture renowned for its proficiency in automatic speech recognition tasks. Ultimately, the overarching aim is to outstrip the performance of SBB's current baseline model.

Given the project's constraints, a conventional approach of manually recording a sufficient number of samples was impractical. Thus, we have been instructed to resort to a pioneering solution: leveraging text-to-speech technology. Exploiting Azure's capabilities, we synthetically generated a large corpus of speech data, an innovative approach that not only addressed the issue of limited data but also introduced a diverse range of accents into the training set.

This report details our journey, presenting our unique methodology of creating a synthetic dataset, our experiences training the Conformer-CTC model, and the subsequent performance evaluation. It unravels our approach to tackling the existing challenges and enhancing the operational effectiveness of SBB's shunting communications, serving as a potential blueprint for similar endeavors in diverse industries.

2 Methods

In the domain of machine learning, especially for tasks that involve speech recognition, the lack of large and diverse datasets is a recurring challenge. This study, therefore, aims to address this gap by employing a novel approach to augment existing datasets with synthetic data, employing a text-to-speech-to-text cycle. We present a comprehensive methodology that is founded on meticulous synthetic data generation and rigorous model training, aimed at creating a reliable speech-to-text model.

Our research primarily revolves around the design and execution of a data pipeline, illustrated in the accompanying diagram, which facilitates synthetic speech data generation, reflecting real-world railway commandos. The pipeline is constituted of three major stages: commandos generation, speech generation, and training set creation. Each stage is specifically engineered to ensure the creation of a high-quality, diverse synthetic dataset.

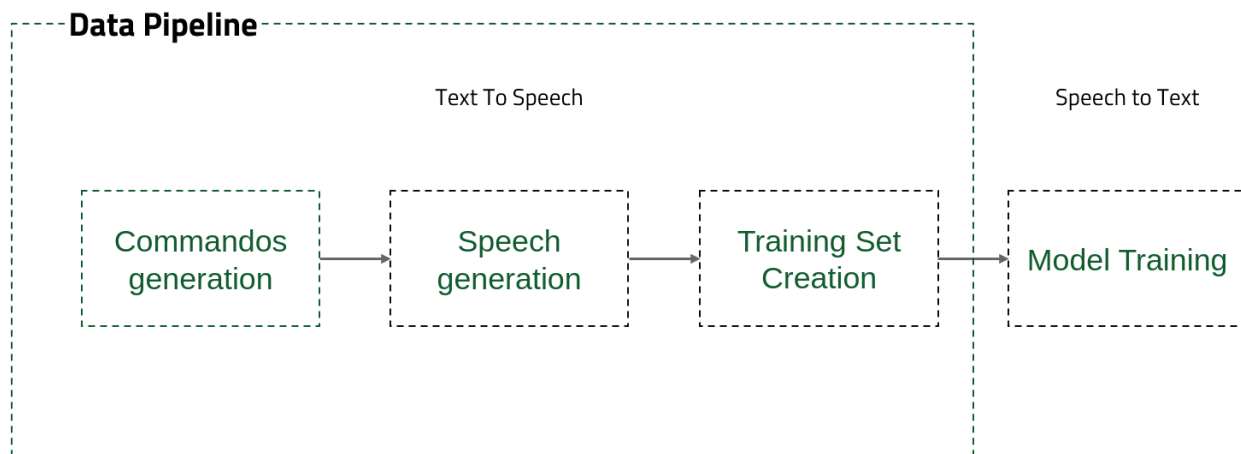


Figure 1: Data Pipeline - Visualization of the Synthetic Data generation steps

2.1 Data Pipeline

The data pipeline forms the crux of our project. Its primary function is to enable the production of synthetic audio data mirroring real-world railway commandos. This aspect significantly enhances the performance of the subsequent speech-to-text model by providing a realistic and diversified training set.

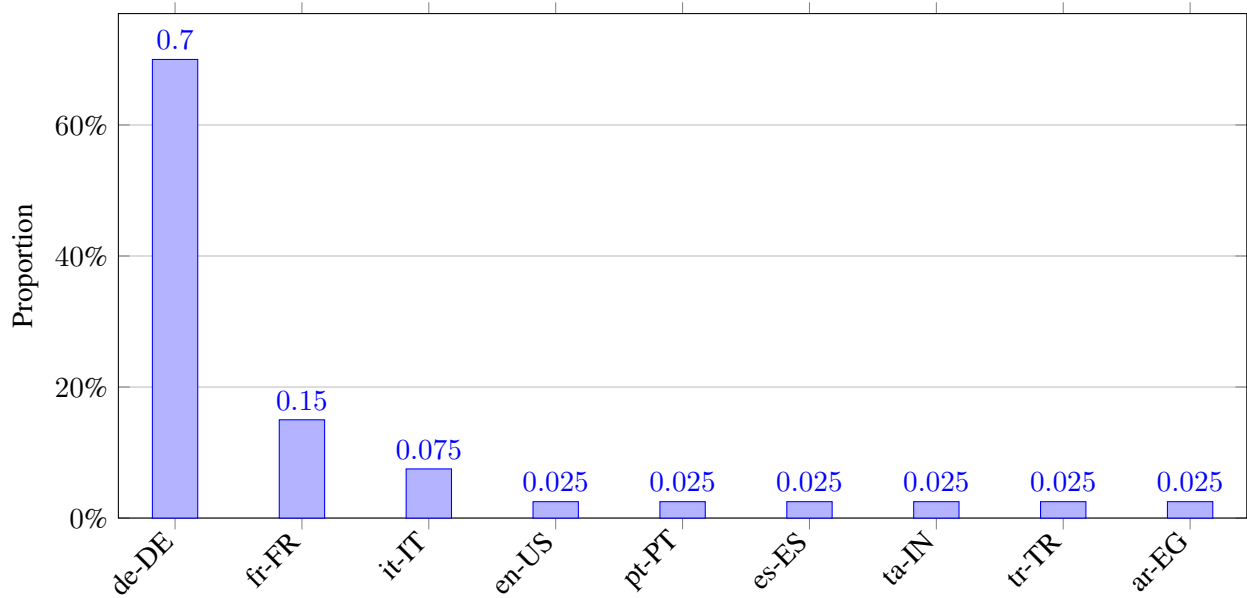


Figure 2: Language distribution of the synthetic samples used in training.

Commandos Generation

The first stage, commandos generation, employs a rule-based system designed to simulate the diversity and unpredictability inherent in real-world commandos. We programmatically generate a vast array of commandos, introducing randomness at multiple intervals. This not only creates a varied dataset but also mimics the true unpredictability of real-life commandos.

Unique to our approach is the integration of guidelines provided by the Swiss Federal Railways (SBB). By incorporating these constraints, our synthetic commandos align closely with actual commands used by railway personnel. This stage represents a delicate balance between following predefined rules and introducing a level of randomness to create a realistic yet diverse set of commandos.

Speech Generation

Upon generating the commandos, they are transformed into speech using Azure’s Software Development Kit (SDK). The choice of Azure SDK was influenced by several factors: its vast library of text-to-speech voices, cost-effectiveness, and compatibility with Speech Synthesis Markup Language (SSML), allowing for fine-grained control over speech parameters.

To add another layer of realism, the parameters for speech generation were carefully selected based on real-world conditions. This includes the gender ratio of SBB’s workforce and the rich linguistic diversity in Switzerland, which comprises German, French, and Italian speakers among others. Thus, the synthetic data generated from our pipeline can mirror the broad range of voices and accents that might be encountered in a real-world setting, enhancing the generalizability of our model.

Training Set Creation

The final phase of the data pipeline revolves around assembling the generated speech samples into a coherent training set for our model. For this purpose, a manifest.json file is created that encapsulates key attributes for each sample. Special care was taken to implement error handling mechanisms during the creation of the manifest.json file, ensuring the consistency and integrity of the training data. We also created a specialized version of the manifest file that includes only samples with German accents, which will facilitate a performance comparison against a broader range of accents.

2.2 Model Training

Model Selection: Conformer CTC

The final component of our methodology is model training. We selected the Conformer-CTC model for this task due to its well-documented proficiency in automatic speech recognition. This model was particularly suited for our requirements because of its non-autoregressive nature, and its ability to capture both local and global interactions in speech.

Furthermore, its support for both sub-word level and character level encodings made it highly adaptable to diverse and complex speech data.

The Conformer-CTC is a variant of the original Conformer model that employs Connectionist Temporal Classification (CTC) loss, which streamlines the model and enhances its efficiency. This model, comprising 120 million parameters, is equipped to capture complex patterns in audio data. It accepts single-channel audio data in WAV format with a sampling rate of 16,000 kHz, aligning well with the high-quality audio data typically encountered in speech recognition tasks.¹

Baseline Creation and Iterative Training

We established a baseline by evaluating the pre-trained model on the human-recorded test set without any additional training. Then, we conducted an experiment where only the human-recorded samples were used for training. These samples, totaling 3924, were split into training (70%), validation (15%), and test sets (15%).

In subsequent iterations, we incorporated synthetic speech data into the training set. Synthetic data was added incrementally, ranging from 10% to 100% of the training set, with 10% increments at each step. This iterative process allowed us to evaluate the model's performance at various levels of synthetic data inclusion, which could provide insights into the optimal amount of synthetic data to use.

Evaluation and Consistency Maintenance

We used the lowest validation word error rate as the criterion to save the best performing model from each iteration, as it is a robust metric for measuring the model's ability to accurately transcribe speech to text. Furthermore, we conducted a manual comparison of selected predictions to perform a qualitative analysis of the model's performance.

To maintain consistency across the experiments and to facilitate reproducibility of our results, we maintained a fixed random seed during the shuffling of training data. We logged all our experiments using Wandb logger, enabling us to easily track and compare the performance of different iterations.

Accent-Specific Training and Methodology Summary

Furthermore, we also performed a special set of experiments where we incorporated only German-spoken synthetic samples into the training set. Given that the test samples were all in German, we hypothesized that training the model with foreign-accented synthetic data might improve its generalization to the test set.

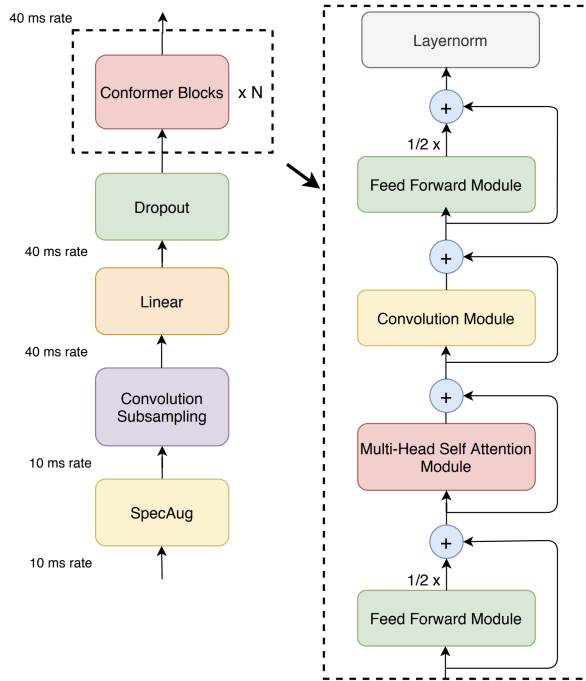


Figure 3: Conformer CTC Architecture

¹Speech Recognition: Conformer CTC." NVIDIA Riva User Guide, Deep Learning Documentation. Available at: <https://docs.nvidia.com/deeplearning/riva/user-guide/docs/reference/models/asr.html>.

To test this hypothesis, we ran additional iterations of the model training, each time incrementally adding german-only synthetic data to the training set for later comparison.

The methodology detailed above provides a comprehensive framework for augmenting existing datasets with synthetic speech data and training a powerful model on this enriched dataset. By comparing the performance of the model when trained with varying amounts of synthetic data and different types of accented voices, we aimed to identify the optimal training strategy to maximize the model’s performance on the task at hand.

3 Results

Our experiments involved training the Conformer-CTC model for automatic speech recognition tasks, focusing on evaluating the influence of synthetic data augmentation on model performance. Performance was measured in terms of Word Error Rate (WER), a standard metric in speech recognition that reflects the rate of transcription errors made by the model.

3.1 Model Performance with Data Augmentation

Data augmentation has been widely adopted as a method of improving model performance in machine learning tasks.² It allows for a larger and more diverse dataset, which is especially crucial in speech recognition tasks where variations in accent, pitch, and speed can significantly impact model performance.

In our case, the introduction of synthetic data was pivotal in expanding the diversity and range of our dataset. Despite the richness of human speech, our dataset comprised a limited variety of combinations regarding speakers and commandos. Many commandos were duplicates, albeit voiced by different speakers. By offering a larger array of text inputs and speaker variations, synthetic data has the potential to improve the model’s generalizability.

	WER (%)
Baseline Conformer CTC pretrained (no training on dataset)	39.48
0% synthetic added (only human samples)	10.07
10% synthetic added	7.69
20% synthetic added	6.64
30% synthetic added	8.27
40% synthetic added	5.40
50% synthetic added	5.44
60% synthetic added	5.50
70% synthetic added	7.28
80% synthetic added	7.03
90% synthetic added	6.49
100% synthetic added	7.60

Table 1: Performance of German only samples with varying levels of synthetic data addition

The baseline Conformer-CTC model, pretrained but without training on our dataset, resulted in a WER of 39.48%. By training the model solely on human-recorded samples we improved the WER to 10.07

As synthetic data was progressively added to the training set, we recorded noteworthy fluctuations in the model’s WER. Notably, the best performance was achieved with the inclusion of 40% synthetic samples, derived from the total available synthetic pool of 16,000 samples, corresponding to approximately 6,400 synthetic samples. This configuration resulted in a training split comprising 25% human-recorded samples and 75% synthetic data, yielding a WER of 5.395%. Beyond this optimal point, the incremental addition of

²Ding et al., "DAGA: Data Augmentation with a Generation Approach for Low-resource Tagging Tasks." EMNLP 2020

synthetic data led to a degradation in performance, although it consistently remained below the baseline of the model trained exclusively with human-recorded samples.

Beyond this optimal point, the incremental addition of synthetic data led to a degradation in performance, although it consistently remained below the baseline of the model trained exclusively with human-recorded samples. Overall, this led to the indication that over-reliance on synthetic data could potentially increase the model’s WER.

This outcome underscores the importance of finding an ideal balance between synthetic and human-recorded data in training. It appears that a considerable amount of synthetic data, about 75% of the total data in this case, can optimize model performance. However, caution must be taken as an over-reliance on synthetic data could potentially increase the model’s WER.

3.2 Accent-Specific Performance

One of the perennial challenges in automatic speech recognition lies in handling various accents. Accents introduce an extra layer of complexity to speech data, resulting in increased variability in pronunciation, intonation, and phonetic features.

With the incremental addition of synthetic data featuring common accents of Switzerland into our training set, we found a slight but manageable degradation in performance. For example, after incorporating 40% of accented synthetic data, the WER achieved was 5.57%, performing similar to the German-only variant. As we increased the percentage of synthetic accented data, we noticed a consistent trend in WER scores:

	German Only (WER %)	Accented Samples (WER %)
10% synthetic added	7.69	6.59
20% synthetic added	6.64	6.40
30% synthetic added	8.27	6.49
40% synthetic added	5.395	5.57
50% synthetic added	5.44	8.47
60% synthetic added	5.50	8.06
70% synthetic added	7.28	7.93
80% synthetic added	7.03	8.17
90% synthetic added	6.49	9.26
100% synthetic added	7.60	9.78

Table 2: Performance comparison: German only vs. Accented Samples

These results are comparable to those obtained with German-only samples, with only a small degradation in performance. This relatively minor performance decrease can be attributed to the fact that the original human-sampled dataset did not include foreign accents, failing to accurately reflect the real-life use case of foreign-accented workers in Switzerland. The Conformer-CTC model was thus faced with the challenge of learning to recognize and transcribe accented speech with limited accented samples.

However, despite this challenge, our model showed robust performance when trained with synthetic accented data. It’s important to highlight that the WER of the models trained with synthetic accented data consistently remained below the baseline model performance, indicating the potential benefits of incorporating synthetic accented data into the training process.

It can be hypothesized that when this model is used in a real-world work setting, the models trained with the inclusion of synthetic accented samples could potentially perform better, as they may generalize better to foreign voices not previously encountered in the training data. The relatively small degradation in performance when introducing accented synthetic data can therefore be interpreted as a promising sign for the practical application of our model.

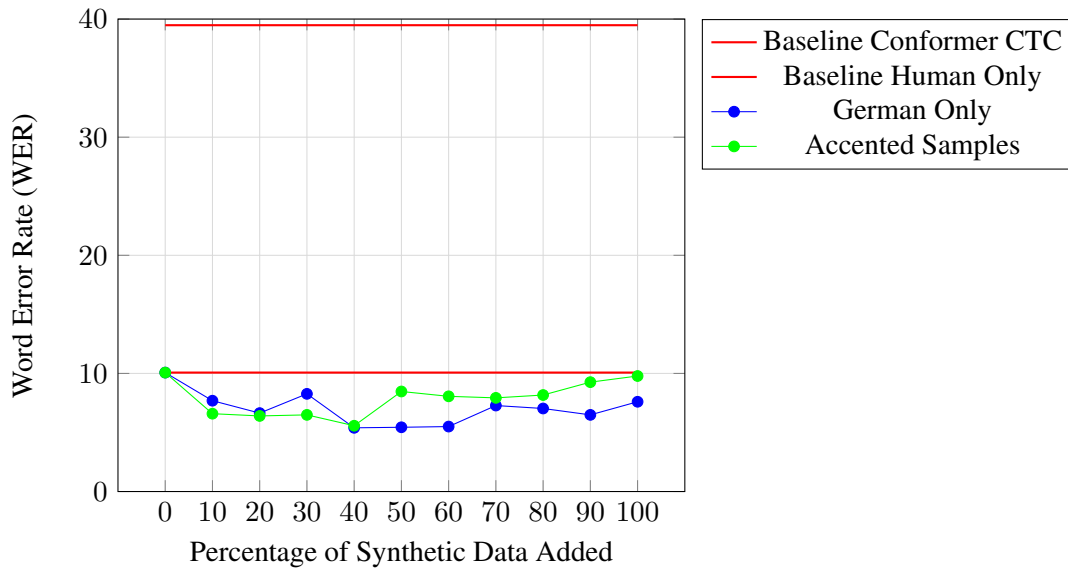


Figure 4: Performance Comparison: German Only vs. Accented Samples

3.3 Qualitative Analysis

To complement the performance metrics, this subsection will provide specific examples of the model’s predictions, highlighting its strengths and weaknesses. This will include instances where the model excelled or struggled in recognizing specific accents or commandos.

The following example shows that our model performs well when transcribing "vier" and "via".

Reference umstellmanöver vom zwei vier via victor
Prediction umstellmanöver vom zwei vier via victor

This indicates that the existing weakness of the SBB model, where "via" and "vier" are sometimes interchanged, could be improved. The assumption is that through the synthetic training data the word „vier" and "via" can be generalized better, because it could be trained with more different speakers. There are only a few exceptions where wrong transcriptions were made, as observed in the following case:

Reference rangierfahrt zwei vier vier golf drei
Prediction rangierfahrt von zwei vier via golf drei

It is noticeable that in many errors, one letter is predicted too much. This concerns especially the following words and predictions:

Reference	rangierfahrt
Prediction	rangierfahrto, rangierfahrvt
Reference	umstellmanöver
Prediction	umstellmanöverv
Reference	xray
Prediction	xrays
Reference	sieben
Prediction	siebena
Reference	whisky
Prediction	whiskey

From our point of view, these incorrect transcriptions can be solved with a simple parser. This would already reduce the WER significantly without having to adapt the model further.

Furthermore, it can be seen that the letter "v" appears again and again in errors:

Reference	umstellmanöver vom
Prediction	umstellmanöverv vonm
Reference	rangierfahrt von fünf
Prediction	rangierfahrt fün fünf
Reference	umstellmanöver vom
Prediction	umstellmanöverv vonm

We suspect that this is related to the pronunciation of the letter. In the German language, this letter is pronounced differently depending on how it is used in the word. This, of course, makes it difficult for the model to predict the letter correctly. How this problem could be addressed is currently unclear.

In conclusion, the quantitative analysis reveals the model's strengths and weaknesses. It performs well in transcribing certain accents and commandos but struggles with interchangeability and occasional letter duplication. We suggest a simple parser to reduce errors without further model adjustments. The recurring issue with the letter "v" highlights pronunciation challenges that need addressing. Overall, the analysis provides valuable insights for enhancing the model's performance.

4 Discussion

Our research has successfully demonstrated the power of synthetic data augmentation in enhancing the performance of a Conformer-CTC model for automatic speech recognition tasks. The notable improvements, particularly with diverse accent inputs, represent a significant step towards overcoming the challenges associated with diverse and accented speech.

In our study, we observed a significant reduction in Word Error Rate compared to the baseline, especially in instances where we incorporated German-accented synthetic samples. This outcome illuminates the potential for synthetic data augmentation to bolster model performance in real-world, multi-accented environments. The Conformer-CTC model's high accuracy, even with limited data, suggests that further improvements could be achieved with more comprehensive and varied training data.

Our findings align with existing research on the effectiveness of data augmentation in speech recognition tasks. However, we extend this understanding by showcasing the practical benefits of synthetic data augmentation in a use case dominated by diverse accents and command variations. This introduces a novel perspective on overcoming the limitations of a confined dataset in automatic speech recognition tasks.

While our research effectively highlights the potential of synthetic data augmentation in improving speech recognition performance, it is not without limitations. Our dataset lacked variation in terms of speakers and commands, indicating room for further improvements. Introduction of a broader range of human voice recordings and command variations could enhance the model's learning and prediction capabilities, fostering a more robust and versatile system.

In addition, despite the promising outcomes, the current model isn't perfect. As evidenced by the qualitative analysis, it can be tripped up by near homophones and other acoustic confusions. However, this could be addressed by future work introducing a second model that parses predictions with the limited classification set provided by the use case, enhancing the model's accuracy.

Given the promising results achieved through the incorporation of synthetic data, future research should explore the use of more sophisticated text-to-speech systems to generate even more realistic synthetic speech. In addition to synthetic data, performance might also be improved by altering existing human samples (e.g., changing their pitch, separating speakers, etc.). Furthermore, future work should explore the performance of specific accents applied to the use case, comparing these to a test set that includes accented speakers to determine if text-to-speech can address the challenges posed by a diverse workforce.

In conclusion, our research illustrates the transformative potential of synthetic data augmentation in advancing automatic speech recognition tasks. Despite the challenges of diverse accents and limited data, we have shown that it's possible to achieve significant improvements using this innovative data augmentation strategy. Our work invites further exploration of synthetic data generation and utilization, which could significantly alter the landscape of automatic speech recognition.

Declaration of Authorship

We declare that this report, and the work presented in it as our own. We confirm that:

- This work was done wholly while in candidature for a degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where we have consulted the published work of others, this is always clearly attributed.
- Where we have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely our own work.
- We have acknowledged all main sources of help.
- Where the thesis is based on work done by ourselves jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signatures:



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